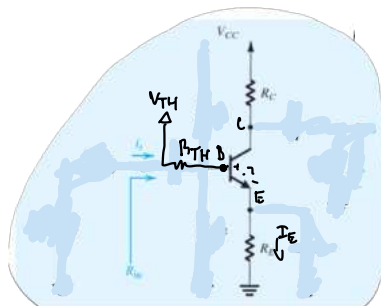
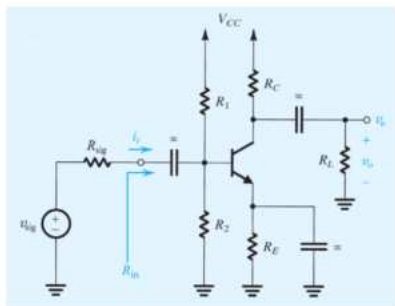


1. For the common-emitter amplifier shown below, let $V_{CC}=15\text{ V}$, $R_1=27\text{ k}\Omega$, $R_2=15\text{ k}\Omega$, $R_E=2.4\text{ k}\Omega$, and $R_C=3.9\text{ k}\Omega$. The transistor has $\beta=100$. Calculate the dc bias current I_C . If the amplifier operates between a source for which $R_{sig}=2\text{ k}\Omega$ and a load of $2\text{ k}\Omega$, replace the transistor with its hybrid- π model, and find the values of R_{in} and the overall voltage gain V_o/V_{sig} .

DC



$$V_{CC} - V_C = I_C R_C$$

$$V_C = V_{CC} - I_C R_C = 7.79$$

$$V_E = (\beta + 1) I_B \cdot R_E = 4.47$$

$$V_C - V_E = 3.31 \approx 2.2 \checkmark$$

$$V_B = V_{TH} = 15 \left(\frac{R_2}{R_1 + R_2} \right) = 5.36 \text{ V}$$

$$R_{TH} = R_1 || R_2 = 9.643 \text{ k}$$

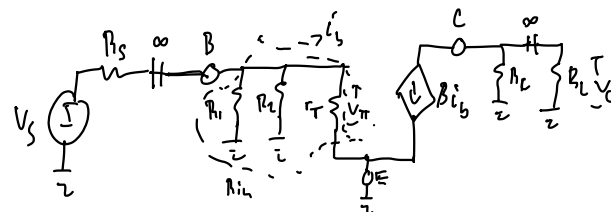
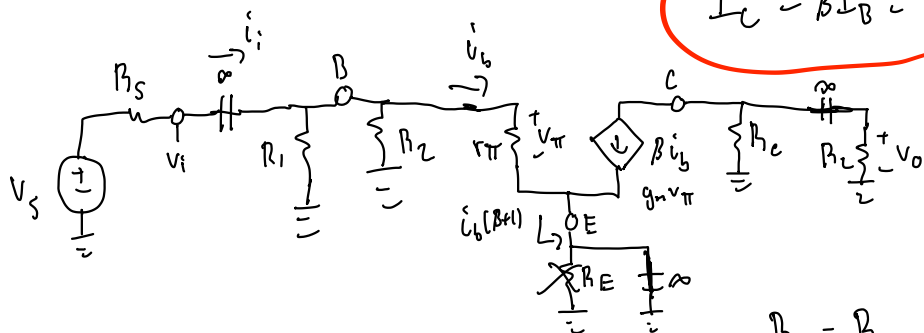
$$V_{TH} - 0.7 = I_B R_{TH} + 0.7 + R_E I_E$$

$$V_{TH} - 0.7 = I_B (R_{TH} + R_E (\beta + 1))$$

$$\frac{4.66}{R_{TH} + R_E (101)} = I_B = 18.478 \mu\text{A}$$

$$I_C = \beta I_B = 1.848 \text{ mA}$$

$$r_{\pi} = \frac{V_T}{I_B} = 1.353 \text{ k}\Omega$$



$$R_{in} = \frac{V_i}{i_i} = R_1 || R_2 || \frac{r_{\pi} i_b}{i_s} \Rightarrow R_{in} = R_{TH} || r_{\pi}$$

$$V_i = V_{sig} \left(\frac{R_{in}}{R_{sig} + R_{in}} \right)$$

$$R_{in} = 1.187 \text{ k}\Omega$$

$$V_o = -\beta i_b (R_C || R_L) = -g_m v_{\pi} (R_C || R_L) \Rightarrow$$

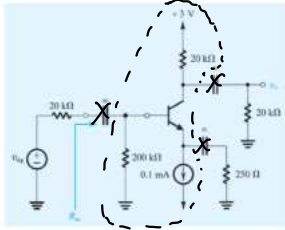
$$V_o = -g_m V_s \left(\frac{R_{in}}{R_{sig} + R_{in}} \right) (R_C || R_L)$$

$$\frac{V_o}{V_s} = -\frac{I_C}{V_T} \left(\frac{R_{in}}{R_{sig} + R_{in}} \right) (R_C || R_L)$$

$$\frac{V_o}{V_s} \approx -36.4$$

2. In the circuit below, the BJT is biased with a constant-current source, and v_{sig} is a small sine-wave signal. Find R_{in} and the gain v_o/v_{sig} . Assume $\beta=100$. If the amplitude of the signal v_{be} is to be limited to 5 mV, what is the largest signal at the input? What is the corresponding signal at the output?

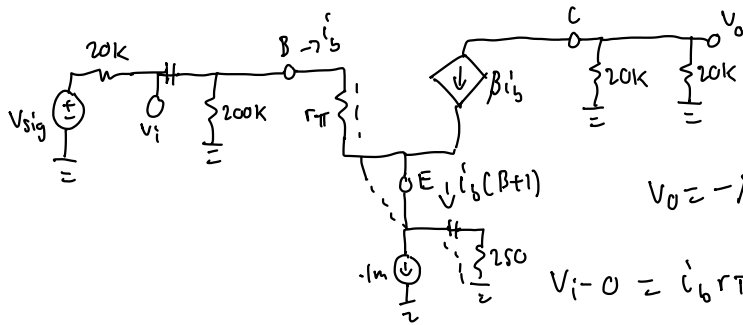
DC



$$I_E = 1 \text{ mA}$$

$$I_B = \frac{I_E}{\beta + 1} = 0.99 \mu\text{A}$$

$$r_{\pi} = \frac{V_T}{I_B} = 2.5, 2.5 \text{ k}\Omega$$



$$v_o = -\beta i_b 10 \text{ k}$$

$$A = \frac{v_o}{v_s} = \frac{v_o}{v_i} \cdot \frac{v_i}{v_s}$$

$$v_i - 0 = i_b r_{\pi} + i_b (\beta + 1) 250$$

$$v_i = i_b (r_{\pi} + 250(\beta + 1))$$

$$v_i = v_s \left(\frac{R_{in}}{R_s + R_{in}} \right)$$

$$A = \frac{-\beta 10 \text{ k}}{r_{\pi} + 250(\beta + 1)} \cdot \frac{R_{in}}{R_s + R_{in}}$$

$$R_{in} = \frac{v_i}{i_i} = 200 \text{ k} \parallel \frac{i_b r_{\pi} + i_b (\beta + 1) 250}{i_b}$$

$$R_{in} = 200 \text{ k} \parallel (r_{\pi} + (\beta + 1) 250)$$

$$R_{in} = 40,321 \text{ k}\Omega$$

$$A \approx -13.26$$

$$v_{be} = v_b - v_e = i_b r_{\pi} = 5 \text{ mV}$$

$$i_b = \frac{5 \text{ mV}}{r_{\pi}} = 1.98 \mu\text{A}$$

$$v_i = v_b \Rightarrow v_b - 0 = i_b r_{\pi} + i_b (\beta + 1) 250$$

$$v_b = 4.44 \text{ mV} \text{ largest input at base signal}$$

$$v_b = v_s \left(\frac{R_{in}}{R_s + R_{in}} \right) = v_s (0.468)$$

$$\frac{v_b}{0.468} \approx v_s \approx 14.44 \text{ mV}$$

largest input signal from source

$$v_o = -\beta i_b 10 \text{ k}$$

$$= -100(10 \text{ k})(1.98 \mu\text{A})$$

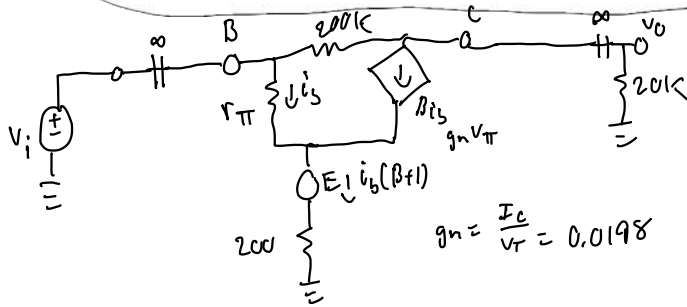
$$v_o \approx -198 \text{ mV}$$

at output

3. The BJT in the circuit below has $\beta=100$.

(a) Find the dc collector current and the dc voltage at the collector.

(b) Replacing the transistor by its π model, draw the small-signal equivalent circuit of the amplifier. Analyze the resulting circuit to determine the voltage gain v_o/v_i .



$$g_m = \frac{I_E}{V_T} = 0.0198$$

$$\text{KCL @ E: } i_e - i_b - g_m v_{\pi} = 0$$

$$\frac{v_e}{200} - \frac{v_b - v_e}{r_{\pi}} - g_m (v_b - v_e) = 0$$

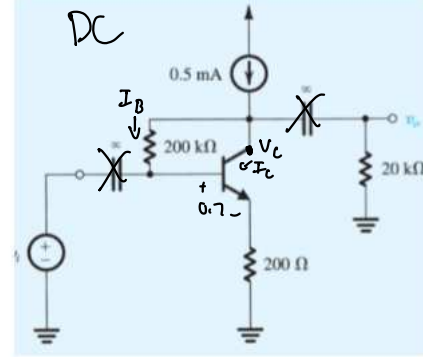
$$v_e \left(\frac{1}{200} + \frac{1}{r_{\pi}} + g_m \right) - v_b \left(\frac{1}{r_{\pi}} + g_m \right) = 0$$

$$\frac{v_e}{v_b} = \frac{v_e}{v_i} = \frac{\frac{1}{r_{\pi}} + g_m}{\left(\frac{1}{200} + \frac{1}{r_{\pi}} + g_m \right)}$$

$$v_e = v_i 0.799 \approx 0.8 v_i$$

$$v_{\pi} = v_b - v_e = v_b (1 - 0.799)$$

$$v_{\pi} = 0.2 v_i$$



$$v_E = 0 = I_E \cdot 200$$

$$v_E = 0.1 \text{ V}$$

$$v_E + 0.7 = v_B$$

$$v_B = 0.8 \text{ V}$$

$$v_C - v_B = I_B \cdot 200 \text{ k}$$

$$v_C = v_B + I_B \cdot 200 \text{ k}$$

$$v_C = 1.79 \text{ V}$$

$$\text{KCL @ } v_C: -0.5 \text{ mA} + I_B + I_C = 0$$

$$I_B (1 + \beta) = 0.5 \text{ mA}$$

$$I_B = \frac{0.5 \text{ mA}}{101} = 4.95 \text{ nA}$$

$$I_C = 0.495 \text{ mA}$$

$$I_E \approx 0.5 \text{ mA}$$

$$r_{\pi} = \frac{V_T}{I_B}$$

$$r_{\pi} = 5.081 \text{ k}\Omega$$

$$\text{KCL @ C: } \frac{v_C}{20 \text{ k}} + \frac{v_C - v_B}{200 \text{ k}} + g_m v_{\pi} = 0$$

$$v_C \left(\frac{1}{20 \text{ k}} + \frac{1}{200 \text{ k}} \right) = \frac{v_b}{200 \text{ k}} - g_m (0.2 v_b)$$

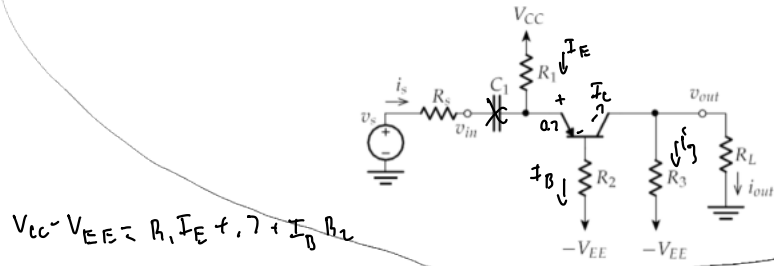
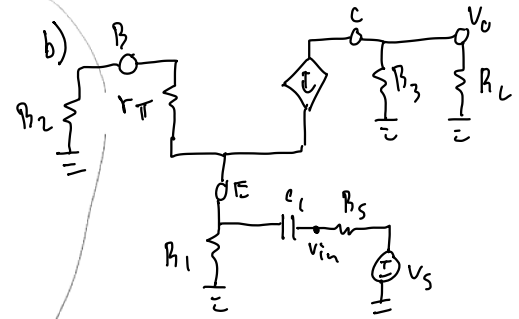
$$v_C \left(\frac{1}{20 \text{ k}} + \frac{1}{200 \text{ k}} \right) = v_i \left(\frac{1}{200 \text{ k}} - 0.00396 \right)$$

$$\frac{v_C}{v_i} = \frac{v_o}{v_i} = \frac{\left(\frac{1}{200 \text{ k}} - 0.00396 \right)}{\left(\frac{1}{20 \text{ k}} + \frac{1}{200 \text{ k}} \right)}$$

$$\frac{v_o}{v_i} \approx -21.91$$

4. For the following circuit, use $R_S = 100 \Omega$, $C_1 = 10 \mu F$, $R_1 = 1 k\Omega$, $R_2 = 220 k\Omega$, $R_3 = 1.5 k\Omega$, $R_L = 500 \Omega$, $V_{CC} = +15 V$, $-V_{EE} = -15 V$, and $\beta = 100$.

- Confirm that the circuit is in the active mode
- Draw the small signal model of the circuit
- Find the amplifier's ac gain (V_{out}/V_{in} with source and load removed)
- Find the small signal input resistance (with the load attached)
- Find the small signal output resistance (with the source attached)
- Sketch the simplified model for the amplifier
- Find the voltage gain of the circuit (with the source and load attached)
- Find the current gain of the circuit (with the source and load attached)



$$V_{CC} - V_{BE} = R_1 I_E + (1 + \beta) I_B R_2$$

$$30 - 0.7 = I_B (R_1 (\beta + 1) + R_2)$$

$$I_B = \frac{29.3}{R_1 + R_2 (\beta + 1)} = 91.3 \mu A$$

$$I_E = (\beta + 1) I_B = 9.22 \text{ mA}$$

$$I_C = \beta I_B = 9.13 \text{ mA}$$

$$V_B - V_{EE} = I_B R_2$$

$$V_B = R_2 I_B + V_{EE}$$

$$V_B = 5.1 V$$

$$V_E = V_B - 0.7 = 5.78 V$$

$$\frac{V_C - V_{EE}}{R_3} + \frac{V_C}{R_L} = I_C$$

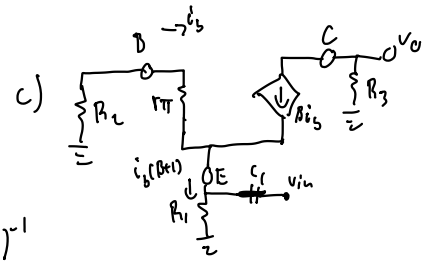
$$V_C \left(\frac{1}{R_3} + \frac{1}{R_L} \right) = I_C + \frac{V_{EE}}{R_3}$$

$$V_C = \left(I_C + \frac{V_{EE}}{R_3} \right) \left(\frac{1}{\frac{1}{R_3} + \frac{1}{R_L}} \right)^{-1}$$

$$V_C = -0.324 V$$

$$V_E - V_C \geq 0.2$$

$$9.12 \geq 0.2 \quad \text{active}$$

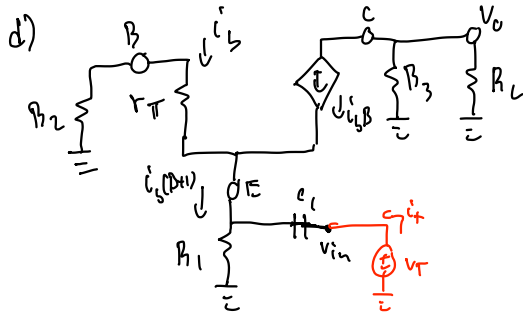


$$V_{in} - 0 = i_b (\beta + 1) R_1$$

$$V_O = -\beta i_b R_3$$

$$\frac{V_O}{V_i} = \frac{-\beta i_b R_3}{i_b R_1 (\beta + 1)} = \frac{-\beta R_3}{R_1 (\beta + 1)}$$

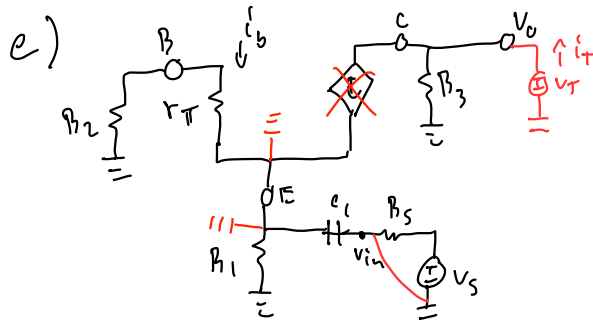
$$\frac{V_O}{V_i} = -1.485$$



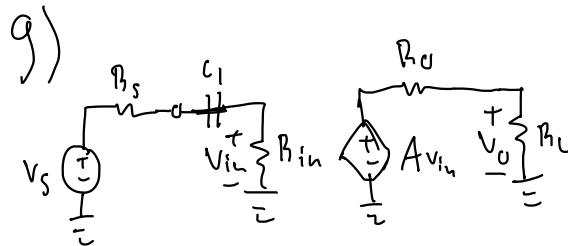
$$R_{in} = \frac{V_i}{i_T} = R_1 \parallel \frac{-i_b (R_2 + R_{in})}{-i_b (\beta + 1)}$$

$$R_{in} = R_1 \parallel \frac{R_2 + R_{in}}{(\beta + 1)}$$

$$R_{in} \approx 685.6 \Omega$$



$$R_{OZ} = \frac{V_T}{i_T} = R_3$$



$$V_{in} = V_S \left(\frac{R_{in}}{R_S + R_{in}} \right)$$

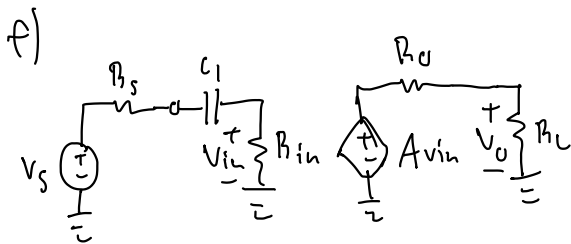
$$V_O = A V_{in} \left(\frac{R_L}{R_O + R_L} \right)$$

$$V_O = \frac{R_L}{R_O + R_L} A V_S \frac{R_{in}}{R_S + R_{in}}$$

$$\frac{V_O}{V_S} = -0.324$$

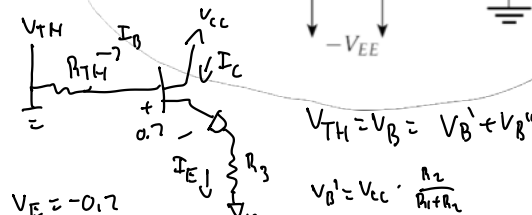
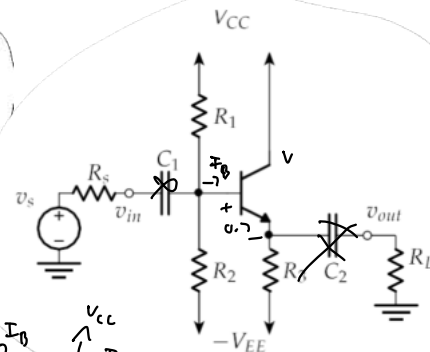
$$A_i = \frac{i_L}{i_{in}} = \frac{V_O}{R_L} \frac{R_S + R_{in}}{V_S} = \frac{V_O}{V_S} \frac{R_S + R_{in}}{R_L}$$

$$A_i = -0.513$$



5. For the following circuit, use $R_S = 500 \Omega$, $C_1 = 0.1 \mu F$, $C_2 = 1 \mu F$, $R_1 = R_2 = 50 k\Omega$, $R_3 = 1 k\Omega$, and $R_L = 100 \Omega$, $V_{CC} = +12 V$, $-V_{EE} = -12 V$, and $\beta = 100$.

- Confirm that the circuit is in the active mode
- Draw the small signal model of the circuit
- Find the amplifier's ac gain (V_{out}/V_{in} with source and load removed)
- Find the small signal input resistance (with the load attached)
- Find the small signal output resistance (with the source attached)
- Sketch the simplified model for the amplifier
- Find the gain of the circuit
- Sketch the Bode magnitude plot



$$V_{TH} = V_B = V_B' + V_B''$$

$$V_B' = V_{CC} \cdot \frac{R_2}{R_1 + R_2}$$

$$V_B'' = V_{EE} \cdot \frac{R_1}{R_1 + R_2}$$

$$V_{TH} = \frac{V_{CC} R_2 + V_{EE} R_1}{R_1 + R_2} = \frac{15(50k) - 12(50k)}{100k}$$

$$V_{TH} = 0$$

$$R_{TH} = 25k$$

$$V_E = -0.7$$

$$0 - V_{EE} = R_{TH} I_B + I_E R_3$$

$$-12 - V_{EE} = I_B (R_{TH} + R_3 (\beta + 1))$$

$$\frac{11.3}{R_{TH} + R_3 (\beta + 1)} = I_B$$

$$I_B = 89.68 \mu A$$

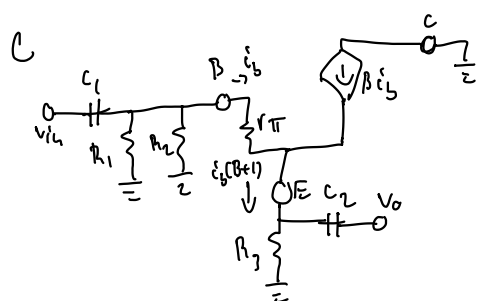
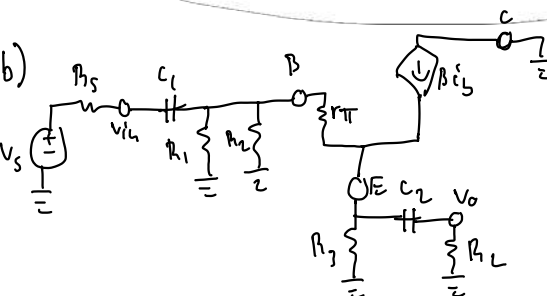
$$I_C = \beta I_B = 8.97 mA$$

$$I_E = (\beta + 1) I_B = 9.1 mA$$

$$r_{\pi} = 278.74 \Omega$$

$$V_C - V_E \geq 0.2 V$$

$$12 - (-0.7) = 12.7 \geq 0.2 \text{ active}$$

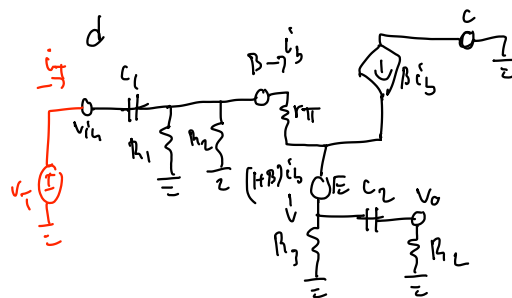


$$v_{in} = i_b r_{\pi} + i_b (\beta + 1) R_3$$

$$v_o = R_3 i_b (\beta + 1)$$

$$A = \frac{v_o}{v_i} = \frac{i_b R_3 (\beta + 1)}{i_b (r_{\pi} + (\beta + 1) R_3)}$$

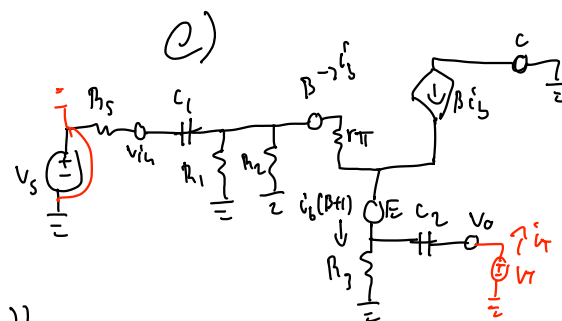
$$A = 0.997$$



$$R_{in} = \frac{v_i}{i_i} = R_1 || R_2 || \frac{i_b (r_{\pi} + (1 + \beta)(R_3 || R_L))}{i_b}$$

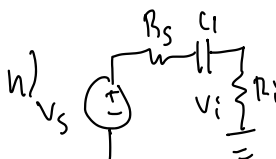
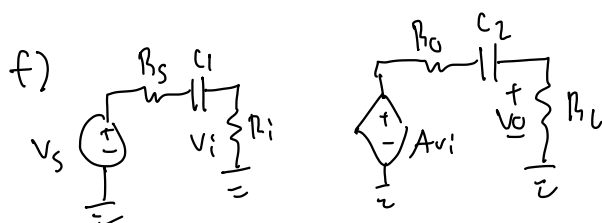
$$R_{in} = R_1 || R_2 || r_{\pi} + (1 + \beta)(R_3 || R_L)$$

$$R_{in} = 6.863 k\Omega$$



$$R_o = \frac{v_i}{i_i} = R_3 || \frac{-i_b (R_S || R_1 || R_2 + r_{\pi})}{i_b (\beta + 1)}$$

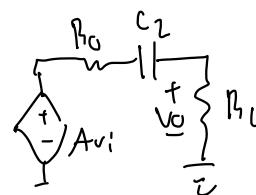
$$R_o \approx 7.56 \Omega$$



$$R_1 = R_S + R_i$$

$$P_1 = \frac{1}{C_1 R_1}$$

$$= 1358.1 \text{ Hz}$$



$$R_2 = R_o + R_L$$

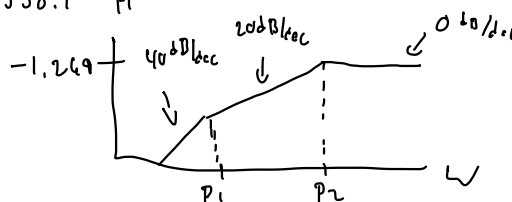
$$P_2 = \frac{1}{C_2 R_2}$$

$$= 9297.5 \text{ Hz}$$

$$g) v_i = V_S \left(\frac{R_i}{R_S + R_i} \right) v_o = A v_i \frac{R_L}{R_L + R_o}$$

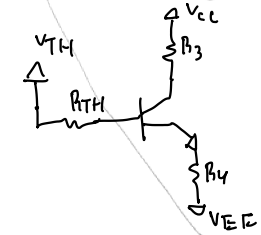
$$v_o = A V_S \frac{R_i}{R_S + R_i} \left(\frac{R_L}{R_L + R_o} \right)$$

$$\frac{v_o}{V_S} \approx 0.864$$



6. For the following circuit, use $R_S = 1 \text{ k}\Omega$, $C_1 = C_2 = 1 \mu\text{F}$, $R_1 = 4.7 \text{ k}\Omega$, $R_2 = 2.2 \text{ k}\Omega$, $R_3 = 1 \text{ k}\Omega$, $R_4 = 1 \text{ k}\Omega$, $R_L = 500 \Omega$, $V_{CC} = +15 \text{ V}$, $-V_{EE} = -15 \text{ V}$, and $\beta = 100$.

- Confirm that the circuit is in the active mode
- Draw the small signal model of the circuit
- Find the amplifier's ac gain (v_{out}/v_{in} with source and load removed)
- Find the small signal input resistance (with the load attached)
- Find the small signal output resistance (with the source attached)
- Find the gain of the circuit
- Using the simplified amplifier model, sketch the Bode magnitude plot



$$V_{TH} = V_T' + V_T''$$

$$V_T' = V_{CC} \left(\frac{R_2}{R_1 + R_2} \right)$$

$$V_T'' = V_{EE} \left(\frac{R_1}{R_3 + R_4} \right)$$

$$V_{TH} = \frac{V_{CC} R_2 + V_{EE} R_1}{R_1 + R_2}$$

$$V_{TH} = \frac{15(2.2\text{k}) - 15(4.7\text{k})}{2.2\text{k} + 4.7\text{k}}$$

$$= -5.435$$

$$R_{TH} = R_1 || R_2$$

$$= 1.499 \text{ k}\Omega$$

$$V_{TH} - V_{EE} = I_B R_{TH} + V_{BE} + I_E R_4$$

$$-5.435 - (-15) = I_B (1.499\text{k}) + 0.7 + I_E (1\text{k})$$

$$I_B = \frac{8.56}{R_{TH} + R_4(\beta+1)} = 86.5 \mu\text{A}$$

$$I_C = 8.65 \text{ mA}$$

$$V_{CC} - V_C = I_C R_3$$

$$V_C = V_{CC} - I_C R_3$$

$$V_C = 9.35 \text{ V}$$

$$V_C - V_E \geq 0.2$$

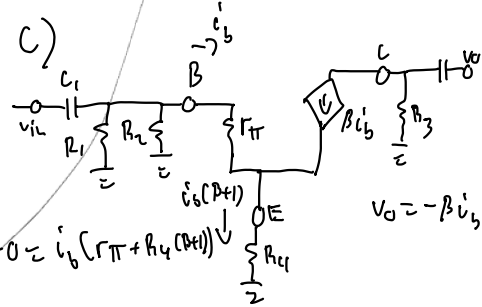
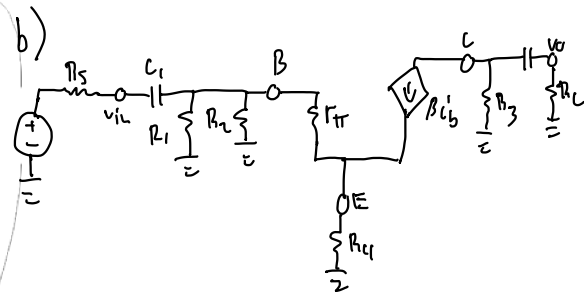
$$12.42 \geq 0.2 \checkmark$$

$$V_{TH} - V_B = I_B R_{TH}$$

$$V_B = V_{TH} - I_B R_{TH}$$

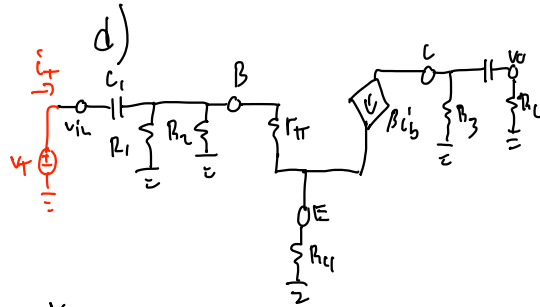
$$V_B = -5.56$$

$$V_E = -6.26$$



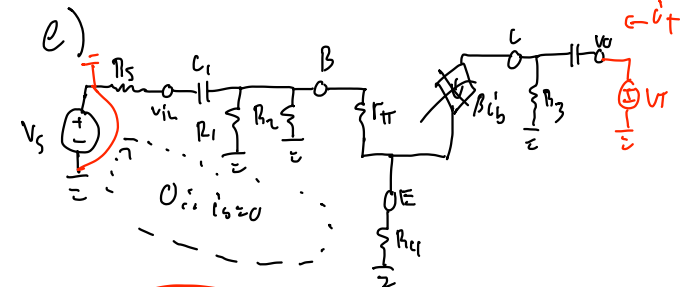
$$\frac{v_o}{v_i} = A = \frac{-\beta R_3}{r_{\pi} + R_4(\beta+1)}$$

$$= -0.987$$

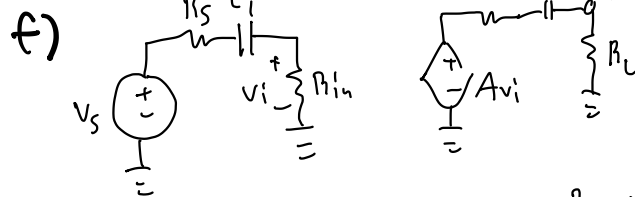


$$R_{in} = \frac{V_T}{i_T} = R_1 || R_2 || r_{\pi} + (\beta+1)R_4$$

$$R_{in} = 1474.7 \Omega$$



$$R_o = \frac{V_T}{i_T} = R_3$$



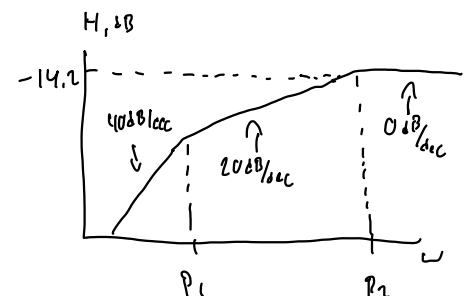
$$v_i = v_s \left(\frac{R_{in}}{R_S + R_{in}} \right) \quad v_o = A v_i \left(\frac{R_L}{R_o + R_L} \right)$$

$$v_o = A v_s \left(\frac{R_{in}}{R_S + R_{in}} \right) \left(\frac{R_L}{R_o + R_L} \right)$$

$$\frac{v_o}{v_s} = -0.146$$

$$g) P_1 = \frac{1}{C_1 (R_S + R_{in})} = 403.74$$

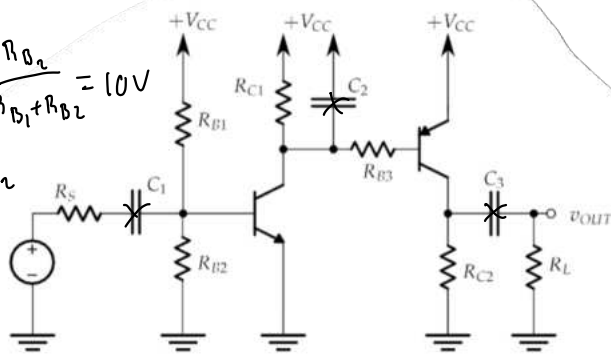
$$P_2 = \frac{1}{C_2 (R_L + R_o)} = 666.66$$



$$V_{TH} = V_{CC} \frac{R_{B2}}{R_{B1} + R_{B2}} = 10V$$

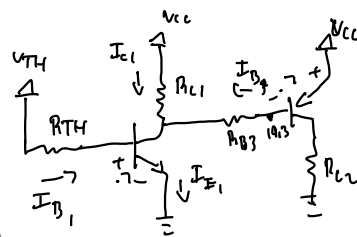
$$R_{TH} = R_{B1} || R_{B2}$$

$$= 50K$$



- (a) Complete a DC analysis to find the values of the small signal model and ensure that the transistor is in an active mode using $V_{CC} = 20V$, $R_S = 10\Omega$, $R_{B1} = R_{B2} = R_{B3} = 100K\Omega$, $R_{C1} = 540\Omega$, $R_{C2} = 1075\Omega$, $R_L = 1000\Omega$, $C_1 = 75\mu F$, $C_2 = 186nF$, $C_3 = 48\mu F$, and $\beta = 100$. Use the DC analysis to compute the parameters for the small signal analysis.
- (b) Write an expression (no values!) for the midband gain, v_{out}/v_{in} , with the source and load removed. Note that v_{in} is the node between R_S and C_1 .
- (c) Find the input and output resistances.
- (d) Estimate the RC time constants for each of the stages of the amplifier.
- (e) Sketch the Bode magnitude plot for the circuit.

=?



$$V_{E1} = 0$$

$$V_{B1} = 0.7$$

$$V_{TH} - 0.7 = I_{B1} R_{TH}$$

$$I_{B1} = 184\mu A$$

$$V_{C1} - V_{C1} = I_{C1} R_{C1}$$

$$V_{C1} = V_{CC} - I_{C1} R_{C1}$$

$$V_{C1} = V_{CC} - \beta I_{B1} R_{C1}$$

$$V_{C1} = 9.954 - V_{E1} \geq 0.2 \checkmark$$

$$\frac{V_{B3} - V_{C1}}{R_{B3}} = I_{B3}$$

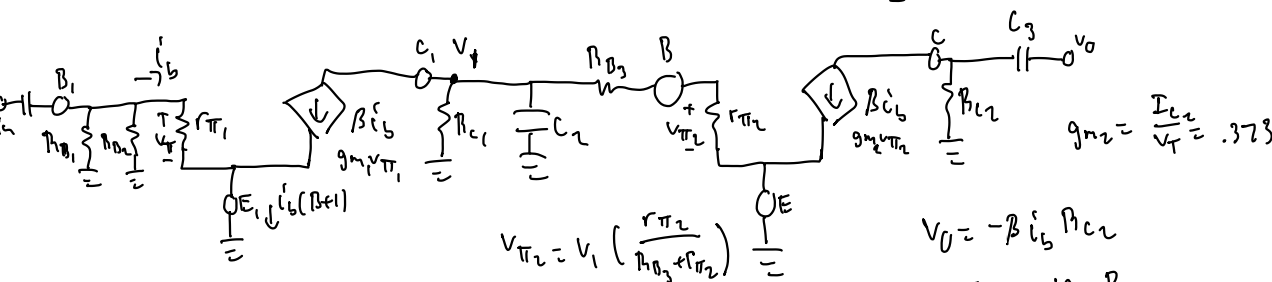
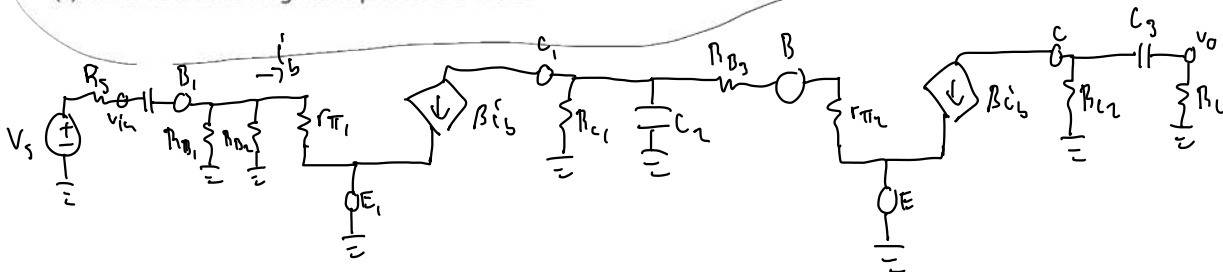
$$I_{B3} = 93.44\mu A$$

$$I_{C3} = 267.55\mu A$$

$$V_{C2} - 0 = R_{C2} I_{C2}$$

$$V_{C2} = R_{C2} \beta I_{B2} = 10.0448V$$

$$V_{E2} - V_{C2} \geq 0.2 \checkmark$$



$$V_{I1} = -\beta i_{B1} R_{C1}$$

$$= -g_{m1} v_{\pi 1} (R_{C1} || (R_{B3} + r_{\pi 2})) = g_{m1} v_{in} (R_{C1} || (R_{B3} + r_{\pi 2}))$$

$$v_{in} = v_{\pi 1} \quad g_{m1} = \frac{I_{C1}}{V_T} = 0.744$$

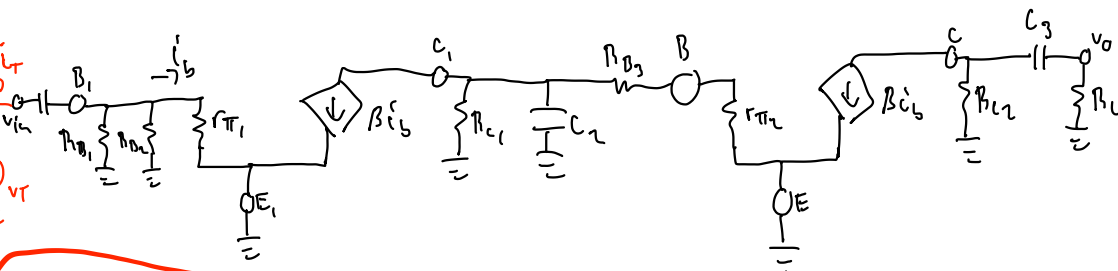
$$V_{O1} = -\beta i_{B1} R_{C1}$$

$$= -g_{m1} v_{\pi 1} R_{C1}$$

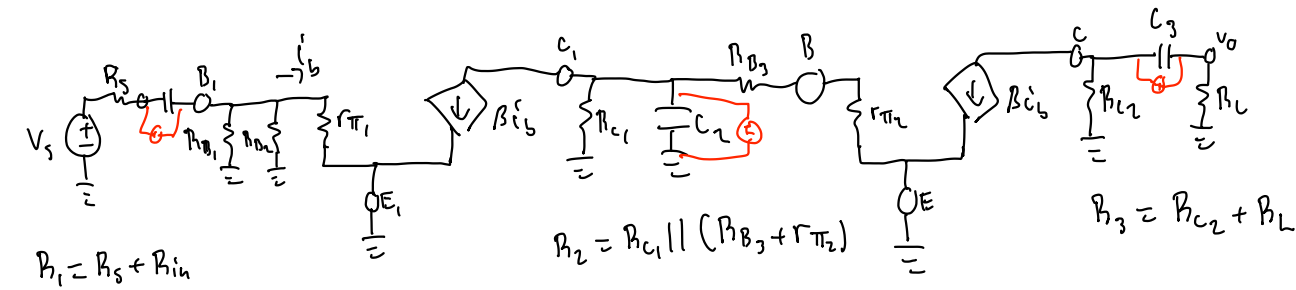
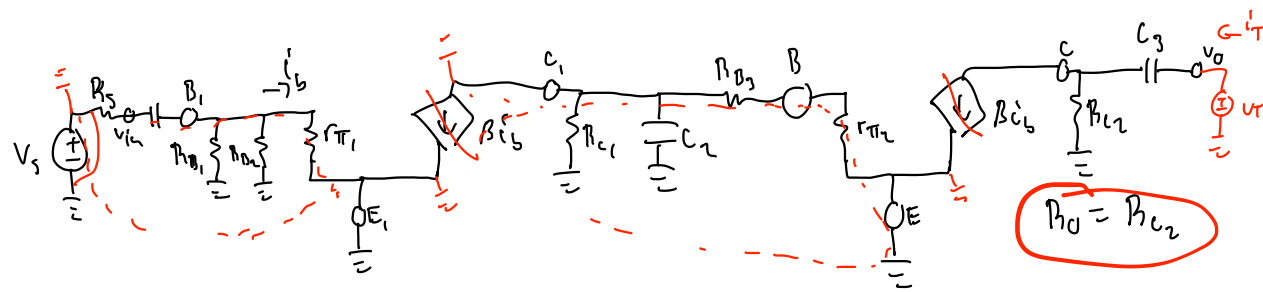
$$= -g_{m1} v_{in} \left(\frac{r_{\pi 2}}{r_{\pi 2} + R_{B3}} \right) R_{C1}$$

$$v_{O1} = -g_{m1} \left(-g_{m2} v_{\pi 2} (R_{C2} || R_L) \right) \left(\frac{r_{\pi 2}}{r_{\pi 2} + R_{B3}} \right) R_{C1}$$

$$\frac{v_{O1}}{v_{in}} = g_{m1} g_{m2} (R_{C1} || (R_{B3} + r_{\pi 2})) R_{C2} \left(\frac{r_{\pi 2}}{r_{\pi 2} + R_{B3}} \right)$$



$$R_{in} = \frac{V_T}{I_T} = R_{B1} || R_{B2} || r_{\pi 1} = 50K || 104K \approx 34.4K$$



$$R_{in} = R_s + R_{in}$$

$$R_{in2} = R_{c1} || (R_{B3} + r_{\pi 2})$$

$$R_3 = R_{c2} + R_L$$

$$\tau = RC = R_1 C_1$$

$$\tau_2 = 49.902 \mu s$$

$$\tau_3 = 44.6 \mu s$$

$$\tau_1 \approx 10.8 \text{ ms}$$

$$p_2 = 10,009.81$$

$$p_3 = 10,04$$

$$p_1 \approx 92,567$$

$$p_3 < p_1 < p_2$$

